

Attendance Regularization Module – Frontend Implementation Guide

Version: 1.0

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Base URL: `/api/v1`

Authentication: Bearer Token (JWT)

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1. Overview

The **Attendance Regularization Module** allows employees to request corrections for their attendance records (e.g., marking an ABSENT day as PRESENT after a valid reason). Managers can approve or reject these requests, and HR Admins can revert previously approved regularizations.

Key Concepts

- **Regularization Request** – A parent request created by an employee for one or more attendance records within a date range.
- **Regularization Detail** – An individual attendance record within a regularization request. Each detail captures the **original state** (snapshot at request time) and the **requested change**.
- **Approval Workflow** – Each detail is individually approved/rejected by the manager. The parent request status auto-updates based on detail outcomes.

2. Authentication

All API calls require a valid JWT token in the `Authorization` header:

`Authorization: Bearer <your-jwt-token>`

The token is obtained from the existing auth/login endpoint.

3. Role-Based Access

Role	Can Do
EMPLOYEE	Create requests, view own requests
MANAGER	View pending requests for subordinates, approve/reject details
HR_ADMIN	Same as MANAGER + revert approved details
SUPER_ADMIN	Same as HR_ADMIN

4. Enums Reference

`AttendanceStatus` (existing – used in attendance records)

Value	Description
-------	-------------

Value	Description
PRESENT	Employee was present
LATE	Employee arrived late
HALF_DAY	Half-day attendance
ABSENT	Employee was absent
LEAVE	On approved leave
HOLIDAY	Company holiday
WEEKEND	Weekend
MISSED_CHECKOUT	Present but forgot to check out

RegularizationStatus (parent request status)

Value	Description
PENDING	All details are pending
PARTIALLY_APPROVED	Some details approved, some still pending
APPROVED	All details approved
REJECTED	All details rejected
CANCELLED	Request cancelled (reserved for future use)

RegularizationDetailStatus (individual detail status)

Value	Description
PENDING	Awaiting manager action
APPROVED	Manager approved the change
REJECTED	Manager rejected the change
REVERTED	Previously approved change was reverted by HR

5. API Endpoints

5.1 Create Regularization Request

Create a new regularization request for one or more attendance records.

POST /api/v1/attendance-regularizations

Access: EMPLOYEE, MANAGER, HR_ADMIN, SUPER_ADMIN

Request Body

```
{
  "fromDate": "2026-08-01",
  "toDate": "2026-08-05",
  "reason": "I was present on these days but my attendance was not recorded",
  "details": [
    {
      "attendanceId": 101,
      "requestedStatus": "PRESENT"
    },
    {
      "attendanceId": 102,
      "requestedStatus": "PRESENT"
    },
    {
      "attendanceId": 103,
      "requestedStatus": "PRESENT"
    }
  ]
}
```

Field Requirements

Field	Type	Required	Description
fromDate	string (date)		Start date of regularization period (YYYY-MM-DD)
toDate	string (date)		End date of regularization period (YYYY-MM-DD)
reason	string		Reason for regularization (non-empty)
details	array		At least one detail item
details[].attendanceId	number (long)		ID of the attendance record to regularize

Field	Type	Required	Description
details[].requestedStatus	AttendanceStatus		The status you want to change to

Response (200 OK)

```
{
  "success": true,
  "message": "Regularization request created successfully.",
  "data": {
    "id": 1,
    "employeeId": 5,
    "employeeName": "John Doe",
    "fromDate": "2026-08-01",
    "toDate": "2026-08-05",
    "reason": "I was present on these days but my attendance was not recorded",
    "status": "PENDING",
    "requestedAt": "2026-08-31T10:30:00",
    "approvedAt": null,
    "approvedById": null,
    "approvedByName": null,
    "rejectedAt": null,
    "rejectedById": null,
    "rejectedByName": null,
    "rejectionReason": null,
    "createdAt": "2026-08-31T10:30:00",
    "details": [
      {
        "id": 1,
        "regularizationId": 1,
        "attendanceId": 101,
        "attendanceDate": "2026-08-01",
        "originalStatus": "ABSENT",
        "originalCheckIn": null,
        "originalCheckOut": null,
        "requestedStatus": "PRESENT",
        "requestedCheckIn": null,
        "requestedCheckOut": null,
        "approvedStatus": null,
        "approvedCheckIn": null,
        "approvedCheckOut": null,
        "status": "PENDING",
        "remarks": null
      }
    ]
  }
}
```

```

    },
    {
      "id": 2,
      "regularizationId": 1,
      "attendanceId": 102,
      "attendanceDate": "2026-08-02",
      "originalStatus": "LATE",
      "originalCheckIn": "10:30",
      "originalCheckOut": "18:00",
      "requestedStatus": "PRESENT",
      "requestedCheckIn": "10:30",
      "requestedCheckOut": "18:00",
      "approvedStatus": null,
      "approvedCheckIn": null,
      "approvedCheckOut": null,
      "status": "PENDING",
      "remarks": null
    },
    {
      "id": 3,
      "regularizationId": 1,
      "attendanceId": 103,
      "attendanceDate": "2026-08-03",
      "originalStatus": "MISSED_CHECKOUT",
      "originalCheckIn": "09:00",
      "originalCheckOut": null,
      "requestedStatus": "PRESENT",
      "requestedCheckIn": "09:00",
      "requestedCheckOut": "18:00",
      "approvedStatus": null,
      "approvedCheckIn": null,
      "approvedCheckOut": null,
      "status": "PENDING",
      "remarks": null
    }
  ]
},
"timestamp": "2026-08-31T10:30:00"
}

```

Error Responses

HTTP Code	Message	When
400	Employee ID is required.	<code>employeeId</code> missing from token resolution
400	At least one detail is required.	Empty details array
400	From date must be before or equal to to date.	Invalid date range
400	Attendance record not found for ID: {id}	Invalid attendanceId
400	Attendance record {id} does not belong to employee {empId}	Attendance belongs to another employee
400	Active regularization request already exists for attendance record {id}	Duplicate request for same attendance
400	Invalid status transition from {current} to {requested}. Allowed: {allowed}	Invalid status change

5.2 Get My Requests

Retrieve all regularization requests submitted by the logged-in employee.

GET /api/v1/attendance-regularizations/my

Access: EMPLOYEE, MANAGER, HR_ADMIN, SUPER_ADMIN

Response (200 OK)

```
{
  "success": true,
  "message": "Regularization requests fetched successfully.",
  "data": [
    {
      "id": 1,
      "employeeId": 5,
      "employeeName": "John Doe",
      "fromDate": "2026-08-01",
      "toDate": "2026-08-05",
      "reason": "Biometric device malfunction",
      "status": "PARTIALLY_APPROVED",
      "requestedAt": "2026-08-31T10:30:00",
      "approvedAt": null,
      "approvedById": null,
    }
  ]
}
```

```
"approvedByName": null,
"rejectedAt": null,
"rejectedById": null,
"rejectedByName": null,
"rejectionReason": null,
"createdAt": "2026-08-31T10:30:00",
"details": [
  {
    "id": 1,
    "regularizationId": 1,
    "attendanceId": 101,
    "attendanceDate": "2026-08-01",
    "originalStatus": "ABSENT",
    "originalCheckIn": null,
    "originalCheckOut": null,
    "requestedStatus": "PRESENT",
    "requestedCheckIn": null,
    "requestedCheckOut": null,
    "approvedStatus": "PRESENT",
    "approvedCheckIn": null,
    "approvedCheckOut": null,
    "status": "APPROVED",
    "remarks": "Verified with security logs"
  },
  {
    "id": 2,
    "regularizationId": 1,
    "attendanceId": 102,
    "attendanceDate": "2026-08-02",
    "originalStatus": "LATE",
    "originalCheckIn": "10:30",
    "originalCheckOut": "18:00",
    "requestedStatus": "PRESENT",
    "requestedCheckIn": "10:30",
    "requestedCheckOut": "18:00",
    "approvedStatus": null,
    "approvedCheckIn": null,
    "approvedCheckOut": null,
    "status": "PENDING",
    "remarks": null
  }
]
}
```



```
"timestamp": "2026-08-31T12:00:00"
}
```

Returns: A list of `RegularizationResponse` objects for the current employee, ordered by creation date (newest first).

5.3 Get Pending Requests (Manager)

Retrieve regularization requests pending approval for the manager's direct subordinates.

GET `/api/v1/attendance-regularizations/pending`

Access: MANAGER, HR_ADMIN, SUPER_ADMIN

Response

Same format as "Get My Requests" – returns a `List<RegularizationResponse>` filtered to requests where:

- The request belongs to a direct subordinate of the logged-in manager
 - The request status is `PENDING` or `PARTIALLY_APPROVED`
-

5.4 Get All Requests (Manager)

Retrieve all regularization requests for the manager's subordinates (any status).

GET `/api/v1/attendance-regularizations/all`

Access: MANAGER, HR_ADMIN, SUPER_ADMIN

Response

Same format as "Get My Requests" – returns a `List<RegularizationResponse>` for all direct subordinates, including pending, approved, rejected, and partially approved requests.

5.5 Get Regularization Details

Retrieve a single regularization request by its ID with all details.

GET `/api/v1/attendance-regularizations/{id}`

Access: EMPLOYEE, MANAGER, HR_ADMIN, SUPER_ADMIN

Path Parameters

Parameter	Type	Description
<code>id</code>	<code>number (long)</code>	Regularization request ID

Response (200 OK)

Returns a single `RegularizationResponse` object with full details.

Error Responses

HTTP Code	Message
404	Regularization request not found with ID: <code>{id}</code>

5.6 Approve/Reject a Detail

Approve or reject an individual regularization detail within a request.

PATCH `/api/v1/attendance-regularizations/{regularizationId}/details/{detailId}`

Access: MANAGER, HR_ADMIN, SUPER_ADMIN

Path Parameters

Parameter	Type	Description
<code>regularizationId</code>	<code>number (long)</code>	Parent regularization request ID
<code>detailId</code>	<code>number (long)</code>	Detail line item ID

Request Body – Approve

```
{
  "status": "APPROVED",
  "approvedStatus": "PRESENT",
  "remarks": "Verified with building access logs"
}
```

Request Body – Reject

```
{
  "status": "REJECTED",
  "approvedStatus": null,
  "remarks": "Unable to verify. Please provide supporting documents."
}
```

Field Requirements

Field	Type	Required	Description
status	Regularization DetailStatus	✓	Must be <code>APPROVED</code> or <code>REJECTED</code>
approvedStatus	AttendanceStatus	⚠ Conditional	The final status to apply to attendance. Required when approving, can be null when rejecting.
remarks	string	✗	Manager's comments

Important: The `approvedStatus` must be the same as or a valid upgrade from the `requestedStatus`. Typically it will match the requested status.

Response (200 OK)

Returns the full updated `RegularizationResponse` with all details (the parent status auto-updates based on detail outcomes).

Error Responses

HTTP Code	Message	When
400	Invalid action. Only <code>APPROVED</code> or <code>REJECTED</code> is allowed.	Status is not <code>APPROVED</code> or <code>REJECTED</code>
400	Regularization detail not found with ID: <code>{detailId}</code>	Invalid detail ID
400	Detail does not belong to regularization request <code>{id}</code>	Mismatched IDs
400	Detail is not in <code>PENDING</code> status. Current status: <code>{status}</code>	Already processed
400	Only the employee's reporting manager can take action.	Unauthorized manager
400	Attendance state has changed since the request was created. Current: <code>{status}</code>	Attendance modified after request

HTTP Code	Message	When
400	Invalid approved status transition from {original} to {approved}.	Invalid status upgrade

5.7 Revert a Detail (HR Admin || Manager)

Revert a previously approved regularization detail back to its original attendance state.

POST /api/v1/attendance-regularizations/{regularizationId}/details/{detailId}/revert

Access: HR_ADMIN, MANAGER, SUPER_ADMIN

Path Parameters

Parameter	Type	Description
regularizationId	number (long)	Parent regularization request ID
detailId	number (long)	Detail line item ID

Request Body

No body required.

Response (200 OK)

```
{
  "success": true,
  "message": "Regularization detail reverted successfully.",
  "data": "Reverted to original attendance state.",
  "timestamp": "2026-08-31T14:00:00"
}
```

What Revert Does

1. Restores the attendance record to its **original snapshot** (status, check-in, check-out at the time the request was created)
2. Changes the detail status from **APPROVED** → **REVERTED**
3. Automatically recalculates the parent request status

Error Responses

HTTP Code	Message	When
400	Detail is not in APPROVED status. Current status: {status}	Only approved details can be reverted
400	Cannot revert: another active regularization request exists for attendance record {id}	Another pending request covers the same attendance

6. Standard Response Format

All endpoints return responses wrapped in a standard envelope:

```
interface ApiResponse<T> {
  success: boolean;      // true if request succeeded
  message: string;       // Human-readable message
  data: T;               // The actual response data
  timestamp: string;     // ISO 8601 timestamp of when the response was
}
```

7. Error Codes

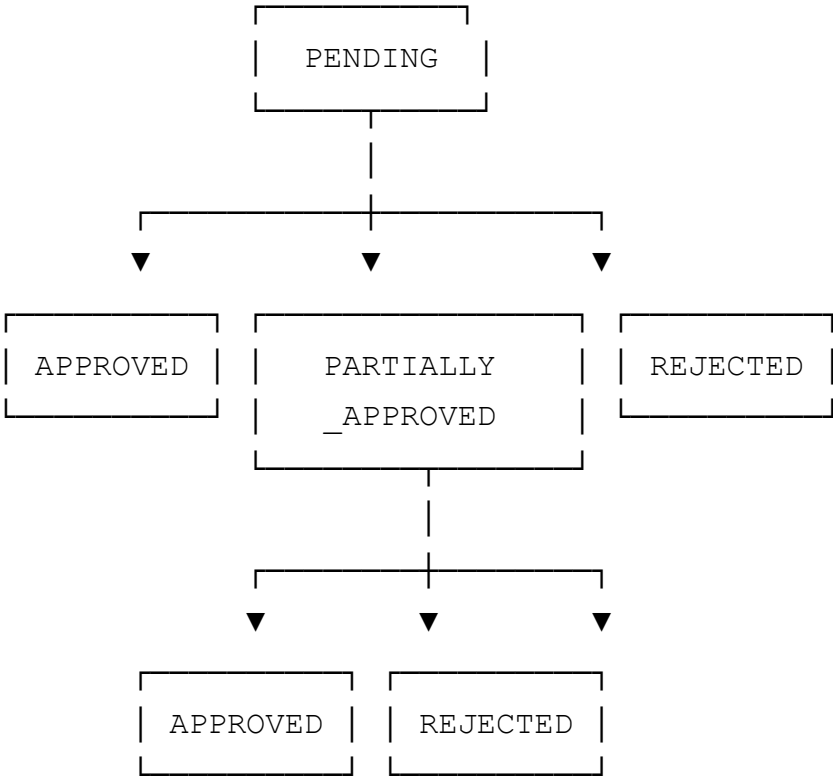
Backend-specific error codes for the regularization module:

Error Code	HTTP Status	Description
REG_001	400	Employee ID is required
REG_002	400	At least one detail is required
REG_003	400	From date must be before or equal to to date
REG_004	404	Regularization request not found
REG_005	400	Active regularization request already exists for attendance record
REG_006	400	Attendance record not found
REG_007	400	Invalid status transition
REG_008	400	Only the reporting manager can take action
REG_009	400	Attendance state has changed since request

Error Code	HTTP Status	Description
REG_010	400	Detail is not in pending status
REG_011	400	Detail is not in approved status (for revert)
REG_012	400	Another active regularization exists (blocks revert)

8. Status Flow Diagrams

Regularization Request (Parent) Status Flow

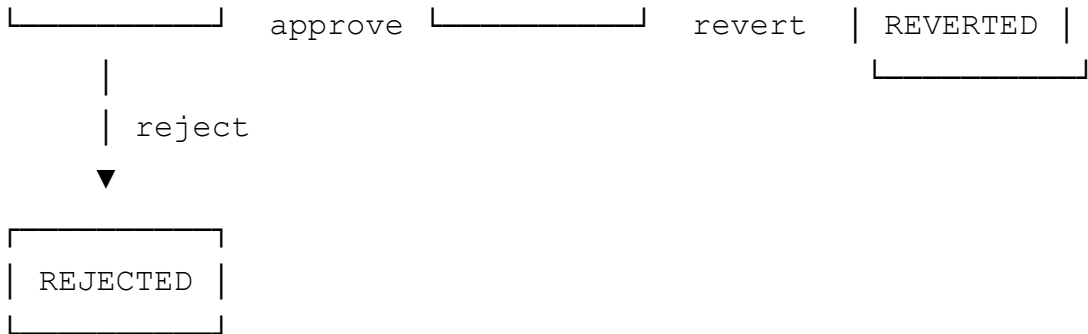


Rules:

- Starts as **PENDING** when created
- Auto-changes to **PARTIALLY_APPROVED** when at least one detail is approved but others are still pending
- Auto-changes to **APPROVED** when all details are approved
- Auto-changes to **REJECTED** when all details are rejected

Regularization Detail Status Flow





Valid Status Transitions (for AttendanceStatus):

Original Status	Allowed Requested Status
LATE	PRESENT
HALF_DAY	PRESENT
ABSENT	PRESENT
MISSED_CHECKOUT	PRESENT
PRESENT	✗ (already present – no change needed)
LEAVE	✗ (not changeable via regularization)
HOLIDAY	✗ (not changeable via regularization)
WEEKEND	✗ (not changeable via regularization)

9. Frontend Implementation Recommendations

9.1 Screens & Components

Screen 1: Employee – My Regularization Requests

Element	Description
Request List Table	Columns: Date Range, Reason, Status (badge), # Details, Created At, Action (View)
Status Filter	Tab/toggle to filter by PENDING, PARTIALLY_APPROVED, APPROVED, REJECTED
Create Button	Opens the "Create Regularization" form/modal
Detail Drawer/Modal	Shows full request details when "View" is clicked

API Calls:

- `GET /my` – Load the employee's regularization requests
 - `GET /{id}` – Load details when a request is clicked
-

Screen 2: Employee – Create Regularization Request

Element	Description
Date Range Picker	From date and To date (or single range picker)
Attendance List	Auto-fetched attendance records within the selected date range. Display as a table with checkboxes.
Status Dropdown	For each selected row, dropdown to choose the requested status (only allowed transitions shown)
Reason Textarea	Free-text reason (required)
Submit Button	Calls <code>POST /</code> endpoint

Workflow:

1. User selects a date range
2. Frontend fetches attendance records for that employee and date range (from the existing attendance module API)
3. User checks the rows they want to regularize and selects the target status
4. User enters a reason
5. Frontend sends `POST /api/v1/attendance-regularizations`

API Calls:

- `GET` existing attendance API (for the employee's records in the date range) – *use existing attendance endpoints*
 - `POST /` – Submit the regularization request
-

Screen 3: Manager – Pending Regularization Requests

Element	Description
Request List Table	Columns: Employee Name, Date Range, Reason, Status, # Pending Details, Created At, Action
Employee Filter	Dropdown to filter by subordinate name
Action Buttons	"Review" to open detail view

Screen 4: Manager – Review & Approve/Reject

Element	Description
Request Header	Employee name, date range, reason, overall status
Details Table	Columns: Date, Original Status, Original In/Out, Requested Status, Requested In/Out, Status, Action
Per-Detail Actions	
– Approve	Button → Opens a small form to confirm <code>approvedStatus</code> and add remarks
– Reject	Button → Opens a form to add rejection remarks (required)
Bulk Actions (optional)	"Approve All" / "Reject All" buttons

API Calls:

- `GET /pending` – Load pending requests
- `GET /{id}` – Load detail view
- `PATCH /{regId}/details/{detailId}` – Approve or reject individual details

Screen 5: HR Admin || Manager – Revert Approval

Same as the Manager review screen, but for `APPROVED` details, shows a "Revert" button instead of Approve/Reject.

API Calls:

- `GET /all` – Load all requests
- `POST /{regId}/details/{detailId}/revert` – Revert an approved detail

9.2 State Management

Recommended state structure (Redux/Zustand/Context):

```
// Regularization state
interface RegularizationState {
  myRequests: RegularizationResponse[];
  pendingRequests: RegularizationResponse[];
  allRequests: RegularizationResponse[];
  selectedRequest: RegularizationResponse | null;
```

```
loading: boolean;
error: string | null;
}
```

9.3 Validation Rules

Client-Side Validation (Create Form):

Rule	Message
From date is required	"Please select a start date"
To date is required	"Please select an end date"
From date ≤ To date	"Start date must be before or equal to end date"
At least one detail selected	"Please select at least one attendance record"
Reason is not empty	"Please provide a reason"
Each detail has a valid requested status	"Please select a target status for each record"

Client-Side Validation (Approve/Reject Form):

Rule	Message
Status is APPROVED or REJECTED	"Please select approve or reject"
If APPROVED: approvedStatus is required	"Please select the final attendance status"
If REJECTED: remarks recommended	"Please add a reason for rejection"

9.4 UI/UX Suggestions

Status Badges

Use color-coded badges for statuses:

Status	Color	Icon
PENDING	 Yellow/Amber	Clock icon
PARTIALLY_APPROVED	 Blue	Partial checkmark
APPROVED	 Green	Checkmark
REJECTED	 Red	X mark
REVERTED	 Orange	Undo icon

Status	Color	Icon
CANCELLED	 Grey	Cancel icon

Request Card (Employee View)

 Regularization Request #1

PENDING

 17

Aug 01, 2026 – Aug 05, 2026

 Biometric device malfunction

 3 attendance records

 Submitted 2 hours ago

Details:

 1 Approved

 2 Pending

 0 Rejected

[View Details →]

Detail Row (Manager View)

 17

Aug 01, 2026

Original:  ABSENT

 | Check-in: -- | Check-out: --

Request:

 PRESENT

 | Check-in: -- | Check-out: --

[ Approve]

[ Reject]

Date Format

All dates should be displayed in a user-friendly format:

- **Display:** MMM DD, YYYY (e.g., "Aug 01, 2026")
- **Time:** hh:mm A (e.g., "10:30 AM")
- **API format:** YYYY-MM-DD (dates) / ISO 8601(timestamps)

Appendix: TypeScript Types

```
// Enums
type AttendanceStatus =
```

```

    | 'PRESENT' | 'LATE' | 'HALF_DAY' | 'ABSENT'
    | 'LEAVE' | 'HOLIDAY' | 'WEEKEND' | 'MISSED_CHECKOUT';

type RegularizationStatus =
    | 'PENDING' | 'PARTIALLY_APPROVED' | 'APPROVED'
    | 'REJECTED' | 'CANCELLED';

type RegularizationDetailStatus =
    | 'PENDING' | 'APPROVED' | 'REJECTED' | 'REVERTED';

// Request Types
interface CreateRegularizationRequest {
    fromDate: string;           // YYYY-MM-DD
    toDate: string;             // YYYY-MM-DD
    reason: string;
    details: CreateRegularizationDetailRequest[];
}

interface CreateRegularizationDetailRequest {
    attendanceId: number;
    requestedStatus: AttendanceStatus;
}

interface RegularizationDetailActionRequest {
    status: 'APPROVED' | 'REJECTED';
    approvedStatus: AttendanceStatus | null;
    remarks?: string;
}

// Response Types
interface RegularizationResponse {
    id: number;
    employeeId: number;
    employeeName: string;
    fromDate: string;
    toDate: string;
    reason: string;
    status: RegularizationStatus;
    requestedAt: string;        // ISO 8601
    approvedAt: string | null;
    approvedById: number | null;
    approvedByName: string | null;
    rejectedAt: string | null;
    rejectedById: number | null;
}

```

```
    rejectedByName: string | null;
    rejectionReason: string | null;
    createdAt: string;
    details: RegularizationDetailResponse[];
}
```

```
interface RegularizationDetailResponse {
    id: number;
    regularizationId: number;
    attendanceId: number;
    attendanceDate: string;
    originalStatus: AttendanceStatus;
    originalCheckIn: string | null;
    originalCheckOut: string | null;
    requestedStatus: AttendanceStatus;
    requestedCheckIn: string | null;
    requestedCheckOut: string | null;
    approvedStatus: AttendanceStatus | null;
    approvedCheckIn: string | null;
    approvedCheckOut: string | null;
    status: RegularizationDetailStatus;
    remarks: string | null;
}
```

```
// Standard API Response
interface ApiResponse<T> {
    success: boolean;
    message: string;
    data: T;
    timestamp: string;
}
```

Appendix: Sample API Call Sequence

Employee Creates a Regularization Request

1. Employee selects date range: Aug 01 – Aug 05
2. [GET] /api/v1/attendance/... (existing endpoint to fetch attendance for that range)
3. Employee sees:
 - Aug 01: ABSENT

- Aug 02: LATE
 - Aug 03: MISSED_CHECKOUT
 - Aug 04: PRESENT (not eligible – already present)
 - Aug 05: HOLIDAY (not eligible – holiday)
4. Employee selects Aug 01, 02, 03 and sets requestedStatus = PRESENT
 5. Employee types reason: "Biometric device was not working"
 6. [POST] /api/v1/attendance-regularizations
 - Returns RegularizationResponse with status = PENDING

Manager Reviews and Approves

1. Manager opens Pending Requests screen
2. [GET] /api/v1/attendance-regularizations/pending
3. Manager clicks on John Doe's request
4. [GET] /api/v1/attendance-regularizations/1
5. Manager sees 3 pending details
6. Manager approves Aug 01:
 - [PATCH] /api/v1/attendance-regularizations/1/details/1
 - Body: { "status": "APPROVED", "approvedStatus": "PRESENT", "remarks": "Verified" }
 - Returns updated response: parent status = PARTIALLY_APPROVED
7. Manager approves Aug 02:
 - [PATCH] /api/v1/attendance-regularizations/1/details/2
 - Returns updated response: parent status = PARTIALLY_APPROVED
8. Manager approves Aug 03:
 - [PATCH] /api/v1/attendance-regularizations/1/details/3
 - Returns updated response: parent status = APPROVED

HR Admin || Manager Reverts an Approval

1. HR Admin opens All Requests screen
2. [GET] /api/v1/attendance-regularizations/all
3. HR Admin finds the approved request and opens it
4. HR Admin clicks "Revert" on Aug 01 detail
5. [POST] /api/v1/attendance-regularizations/1/details/1/revert
 - Attendance for Aug 01 is restored to ABSENT
 - Detail status changes to REVERTED
 - Parent status recalculates to PARTIALLY_APPROVED